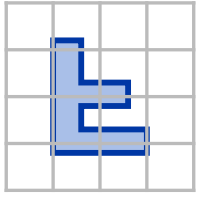
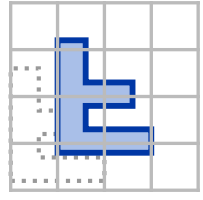
identity



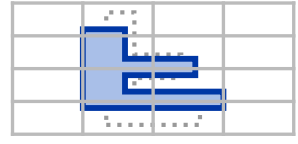
$A = I$

translation



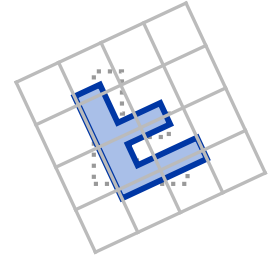
t_x, t_y in last column

scaling



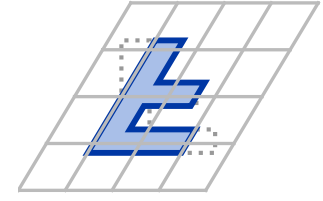
$\mathrm{diag}(a, b, 1)$ = voxel size

rotation



$\cos\alpha, \ \pm\sin\alpha$

shear



off-diagonal c_x